اونيۏرسيتي مليسيا ڤهڠ السلطان عبد الله
UNIVERSITI MALAYSIA PAHANG
AL-SULTAN ABDULLAH

BSD 1323 STORYTELLING AND DATA VISUALISATION
INDIVIDUAL PROJECT

TOPIC:

MONEYBALL: VISUALISING THE FOOTBALL TRANSFER MARKET

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TOPIC MOTIVATION

There are several motives as to why I choose this topic as my domain for this individual project. First of all, it is solely based on my personal interest. Apart from the knowledge of data science, football is also one of the few things that I am really passionate about. As someone who loves numbers and football, utilising the knowledge of data analytics in this topic is something I have been meaning to do. Being a football data analyst is my ultimate dream job once I graduate from my data analytics degree at UMP.

Other than that, I want to study more details about the football transfer market. Specifically in Europe where it is the home of modern football. Business management has been a crucial part in football through the years. Nowadays, a football institution can not be run properly without the presence of money. Hence, there exists the term *"Moneyball"* not just in football, but generally in any sport. The saddest truth is that, in this era, footballers' overall values are often judged based on a certain amount of money, known as the market value. Therefore, I want to find out how factors such as geography, time period and performance can affect a player's value in this sport.

DATASET

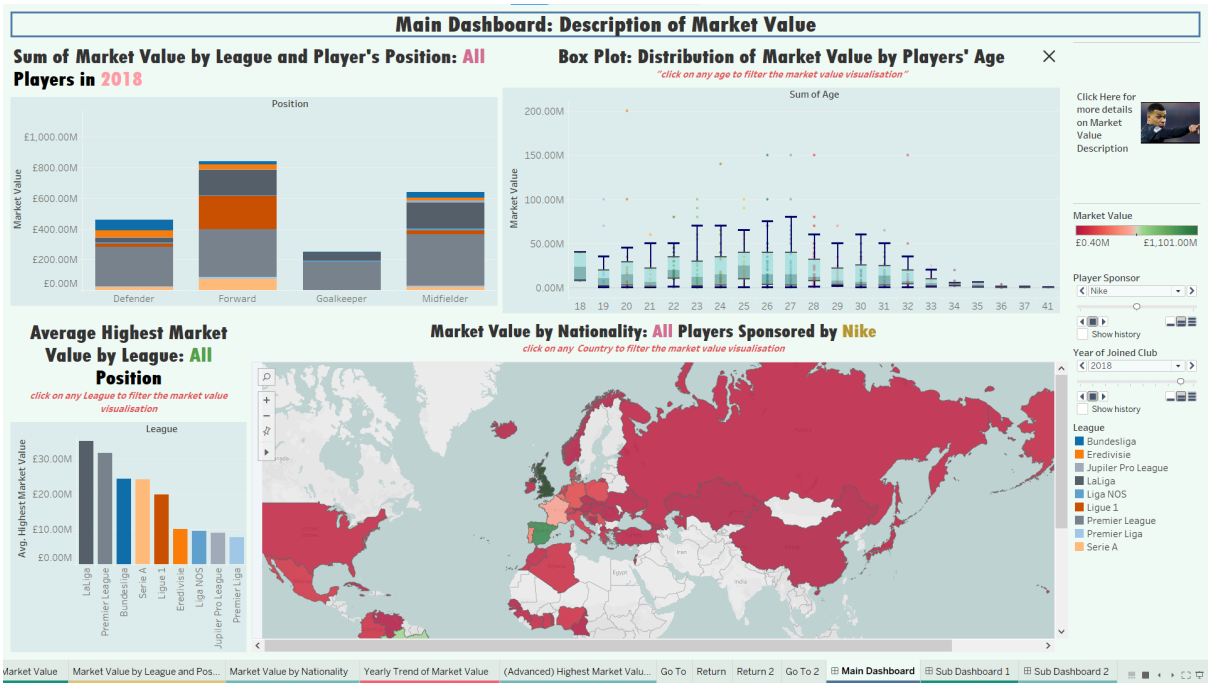
The dataset is obtained from the kaggle website. At first there are over 4000 records, but after removing the null values and filtering it using tableau, there are a total of 516 records. This dataset consists of 11 fields, excluding hidden fields that are not being used in this project. The dataset displays data that are useful to analyse the overall market value of footballers around europe.
The detailed explanation of each fields are as below:

No	Field Name	Data Type	Explanation
1	Age	Numerical	Players' current age in which this data is recorded, specifically in 2018 and 2019.
2	Highest Market Value Date	Date	The date in which a player's market value was the highest.
3	Joined Club	Date	The date when a player joined a club with a certain amount of fee as compared to his current market value.
4	Nationality	Geographic	The player's country of origin.
5	Player Name	Categorical	Player's name.
6	League	Categorical	Players' current league when data is recorded. There are a total of seven leagues in this dataset, which are the top 7 biggest leagues in europe.
7	Player Sponsor	Categorical	Sports brands that are associated with the players.
8	Position	Categorical	Players' position in the team. It can be either goalkeeper, defender, midfielder, or forward.
9	Accumulated Transfer Sums	Numerical	The total transfer fee of a player when moving to another club along his career (in Pound Sterling). E.g: Lionel Messi has never moved to another club since his professional debut, so his transfer sum is £0.

10	Highest Market Value	Numerical	Player’s highest recorded market value along his career (in Pound Sterling)
11	Market Value	Numerical	The Market Value of a Player (in Pound Sterling)

DATA INTERPRETATION AND ANALYSIS

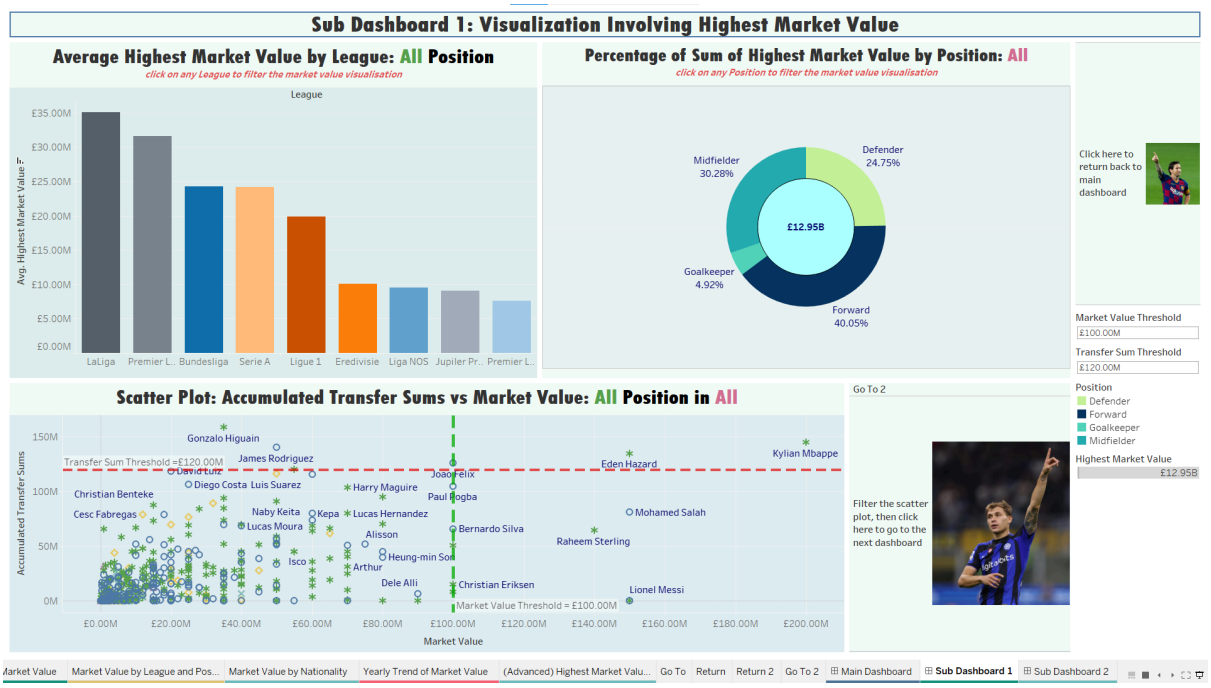
Main Dashboard



Content: Visualisation 1, Visualisation 2, Visualisation 4, Visualisation 5

- The main dashboard mainly focuses on the description of market value, consisting of three visualisations for analysis purposes, and one for filtering purposes only.
- The three main visualisations are sum of market value by position, the box plot, and the heat map.
- In this dashboard, users can filter the age on the box plot, by selecting on any specific age to see the market value by certain players' age only.
- As for the heat map, it is used as a nationality filter. To see each visualisations by a specific country, users can select any country and the menu will be displayed.
- Lastly, the average highest market value bar chart is only used for filtering purposes, filtering can be done by hovering on the chosen league.
- A Go To sheet was also put in this dashboard, by clicking the picture of Kylian Mbappe at the upper right corner, users can move to sub dashboard 1.

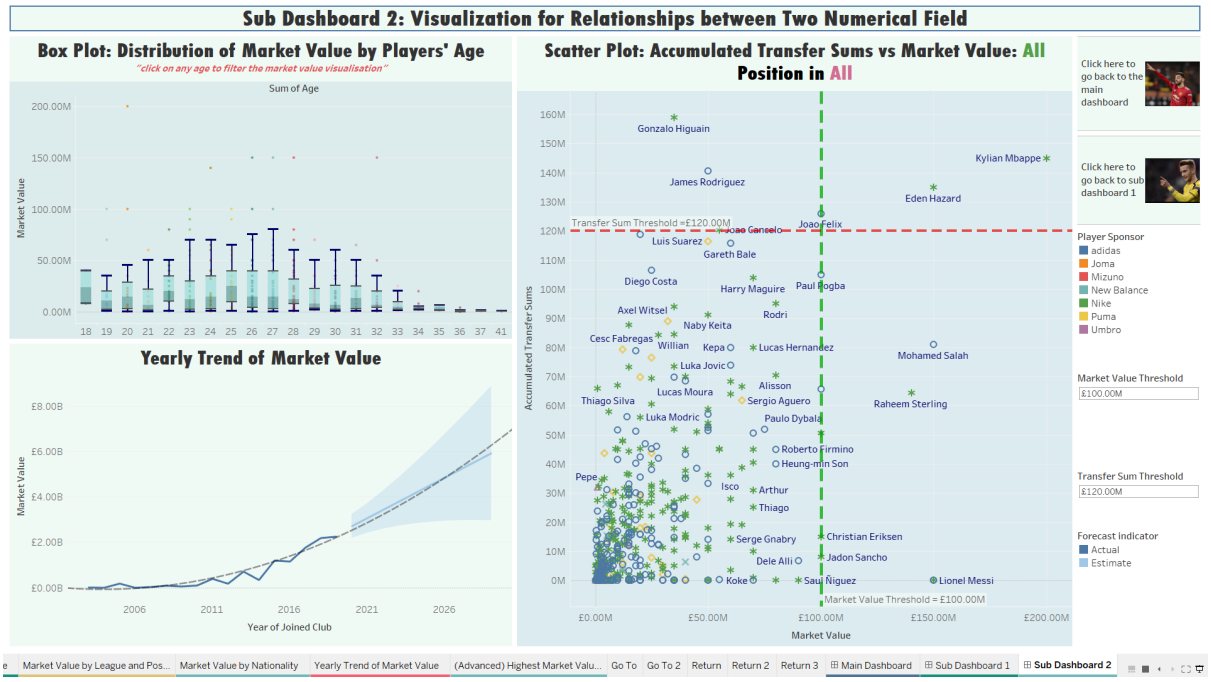
Sub Dashboard 1



Content: Visualisation 1, Visualisation 5, Visualisation 7

- The first sub dashboard specifically shows the visualisations in terms of highest market value of a player.
- This time, the bar chart is used for analysis and filterings, Bar chart can filter the certain league selected by users.
- As for the donut chart, when users select a specific position on the donut chart, other visualisations will display the record for that position only.
- To go to the next sub dashboard, the user must filter the scatter plot first, then click on the picture of Nicolo Barella beside the scatter plot.
- To return to the main dashboard, simply click the icon of Lionel Messi on the upper right corner.

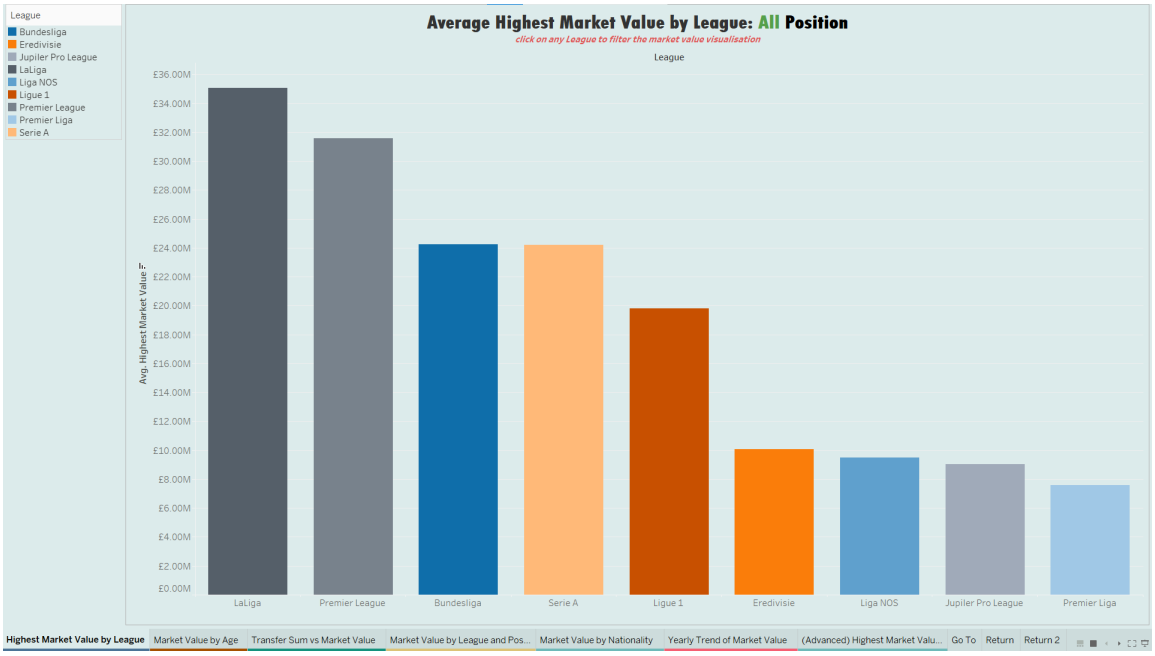
Sub Dashboard 2



Content: Visualisation 2, Visualisation 3, Visualisation 6

- The second dashboard focuses on the relationships between two numerical variables.
- Box Plot distribution sees the market value for all ages of players, indicating the average and variability comparisons.
- It is also used to filter the age just like in the main dashboard.
- The yearly trend of market value shows the total sum of market value over the years, as well as showing the forecast estimate for the market value in the next 10 years.
- This dashboard also allows users to see more clearly on the scatter plot that displays the visualisation of transfer sum vs market value.
- Filtering the scatter plot and seeing it in the bigger size compared to sub dashboard 1 can help users understand more briefly about the relationship.
- To return to the main dashboard, simply click the go to sheet with the picture of Bruno Fernandes on the upper right corner.
- To go to sub dashboard 1, click the picture of Marco Reus.

Visualisation 1

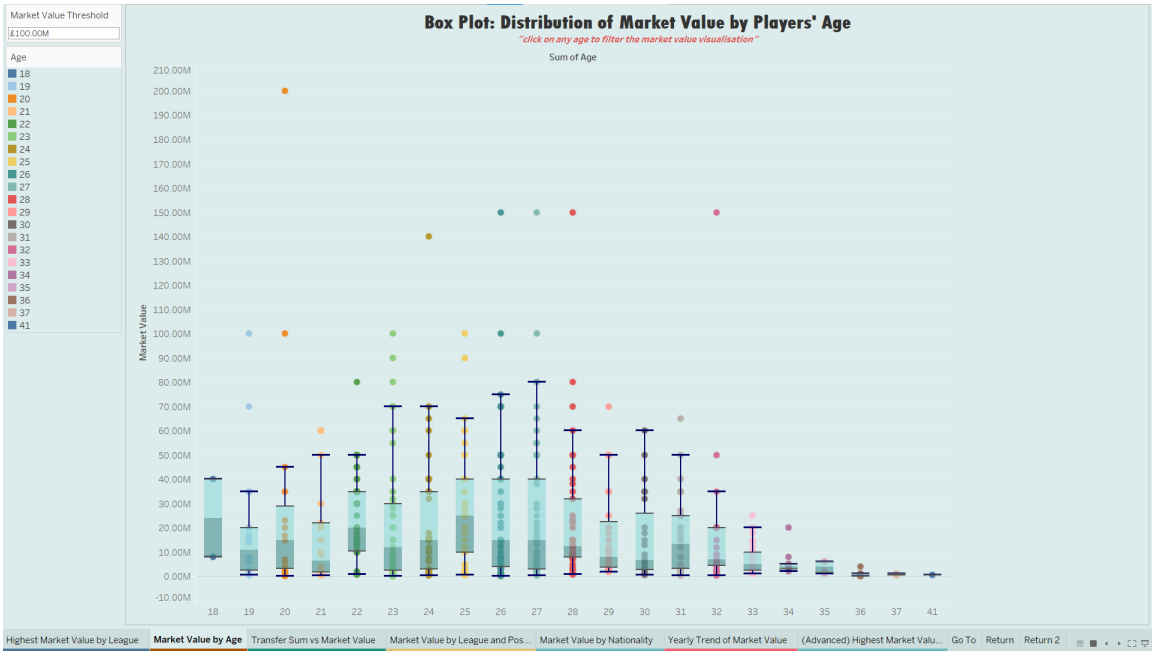


Title: Average Highest Market Value by League: All Position

Chart Type: Bar Chart

- In this visualisation, we can see the comparison between the average highest market value of players from different leagues.
- It can be filtered by the player's position from visualisation 7.
- Overall, we can observe that La liga has the highest average market value among all the leagues, followed closely by the Premier League.
- This is because both leagues are well known for their high value players, most of the world's greatest footballing talent are playing in these leagues.
- Plus, La liga and Premier League are the two most watched football leagues in the world.
- As a data analyst, scouting players from these leagues can be beneficial in finding the perfect talent to strengthen my team.

Visualisation 2

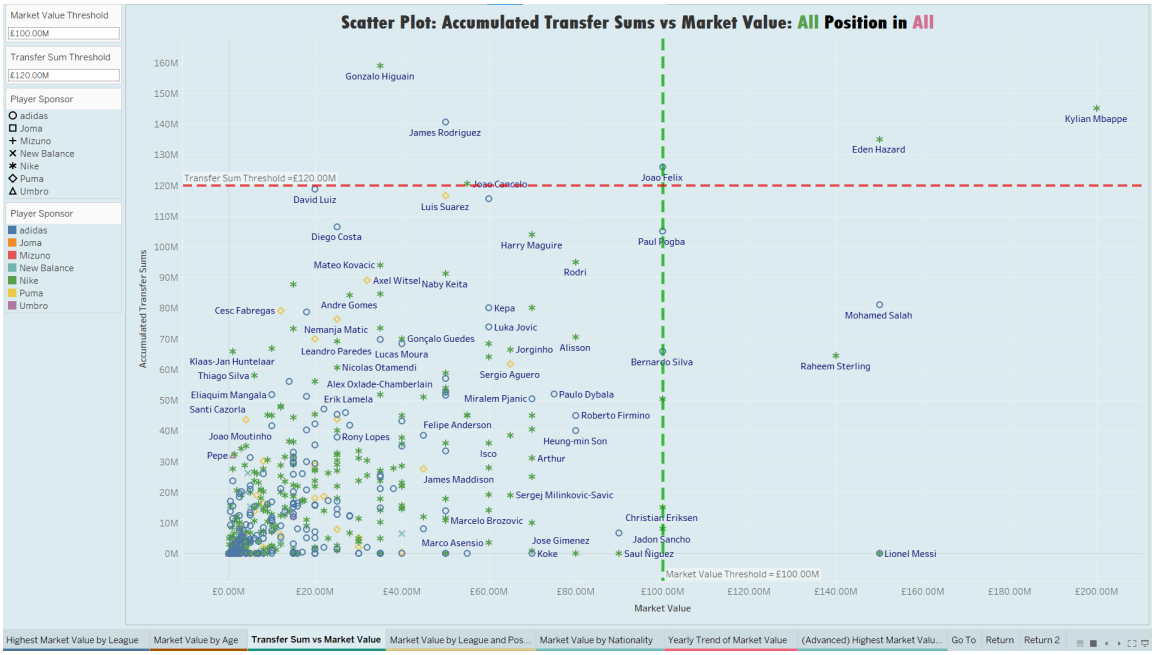


Title: Box Plot: Distribution of Market Value by Player's Age

Chart Type: Box Plot

- We can interpret that every level of ages has a positive skewed distribution since the medians are all nearer to the first quartile.
- Players age 25 have the largest median or average compared to other ages.
- In terms of variability, players age 27 has the most variation since it has the longest interquartile range.
- This is because most footballers' value start to rise and peak at the age of 25 until 27. In football, we call this the *prime year/era*.
- As we can see the median gets relatively lower when it comes to age 32 above, this is the complete opposite of prime years, footballers at this age start to experience their downfall era, or we call this *past their prime*.
- By ignoring outliers, if a football team is searching for great talents for a long term plan, they might need to consider footballers at the age 25-27.
- Otherwise, if they are searching for an experienced squad player and cheaper options, they can target players that have past their prime.

Visualisation 3

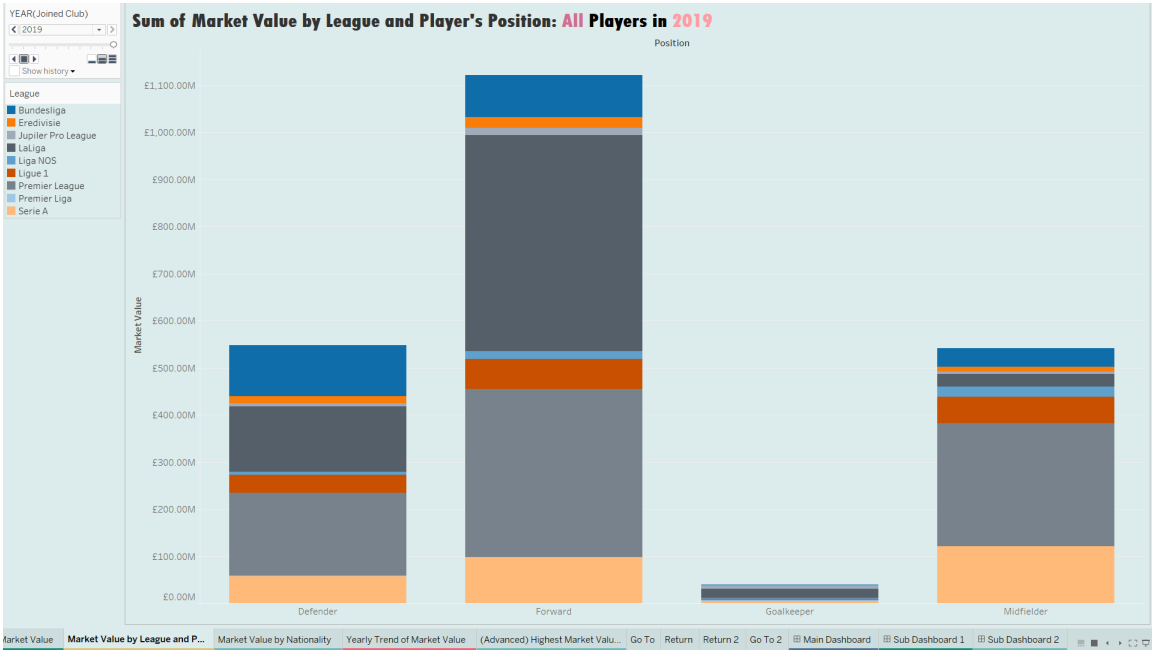


Title: Accumulated Transfer Sums vs Market Value

Chart Type: Scatter Plot

- This visualisation displays the relationship between accumulated transfer sums and market value of players.
- Different colours and shapes both represent different sponsors.
- This can be filtered by its league and position using visualisation
- The transfer sum threshold is set to £120M, while the market value threshold is set to £100M.
- As we can see, the upper right section are the players with high market value and high accumulated transfer sum, these are the most expensive and valuable players in world football.
- On the other side, we also have players that have really high transfer sums but low market value like Gonzalo Higuain. These players are not necessarily top players, but they are just players that always change clubs all the time.
- Sponsors also play a big role in this matter.
- Most players with high market value and transfer sum are sponsored by Adidas and Nike, two of the biggest sports brands in the world.
- A good data analyst will not target any players that exceed both thresholds, unless the club has a big project being planned for the future.

Visualisation 4

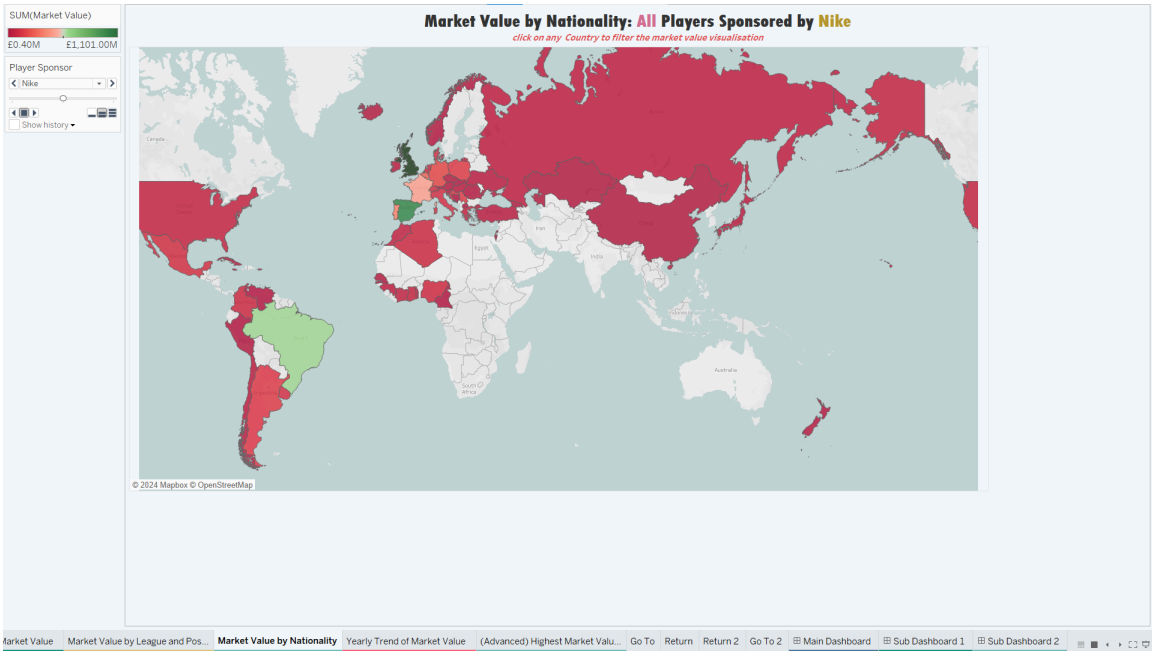


Title: Sum of Market Value by League and Player's Position: Year of Joined Club

Chart Type: Stacked Bar Chart

- The visualisation shows the sum of market value for each position, partitioned by each league.
- The records are divided by pages, which are the year in which the player joined his current club, starting from 2010.
- We can see significant increases for every position, indicating how the overall market value of players gets higher every year.
- Every year, the premier league covers the highest sum of market value for every position.
- This shows how the value of the premier league maintains every year, meaning that it is still the biggest football league in the world since the last 10 years.
- In terms of player's position, forward has the highest sum of market value.
- This is because generally in every team, it is a common perception that forwards are always the star player. Hence the high market value.
- To conclude, football clubs often need to invest more in purchasing forwards as compared to other positions.

Visualisation 5

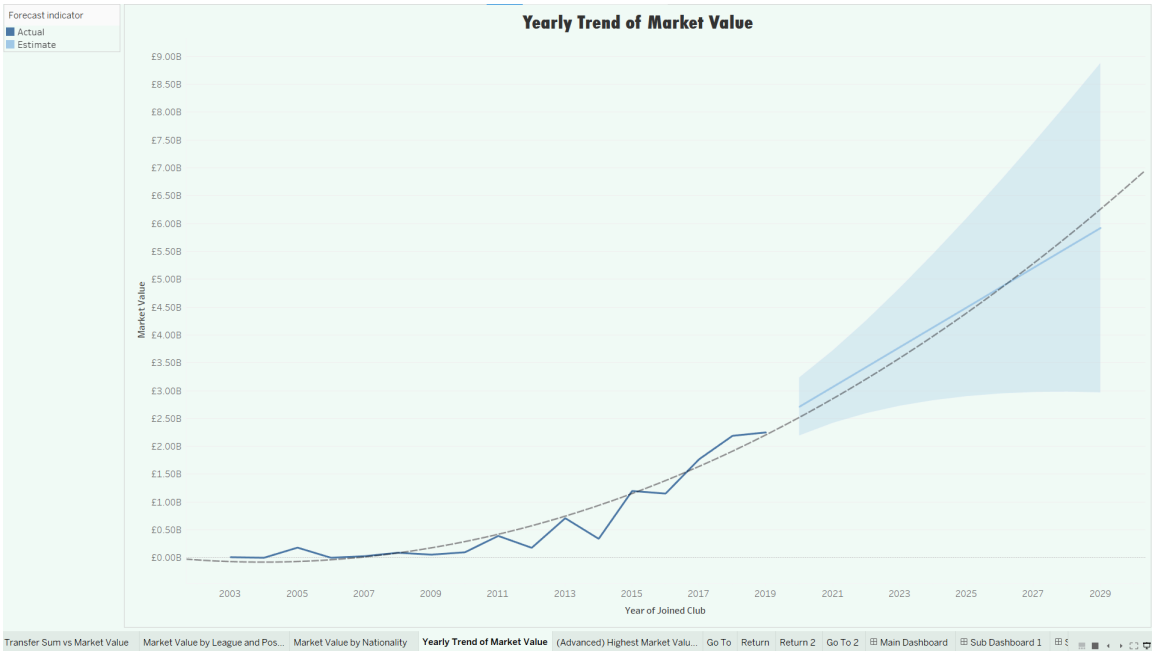


Title: Market Value by Nationality

Chart Type: Heat Map

- This heat map shows the sum of market value by players' country of origin.
- This visualisation can be filtered by league visualisation 1.
- The representation was also divided into several pages namely Nike, Adidas and Puma sponsor only.
- Red colour indicates low market value, green colour shows high market value.
- For Adidas and Puma, Spain has got the highest sum of market value.
- As for Nike, three countries have the obvious high market value with the United Kingdom being the top, followed by Spain, Brazil and France.
- All the countries mentioned have the greatest power in world football. They are the biggest footballing nations with the best football foundation in the world.
- Considering the fact that most players from these countries are sponsored by big brands, their value will definitely rise higher.
- That is why up until today, many football clubs scout players from these four countries and want to develop them to become bigger stars.

Visualisation 6



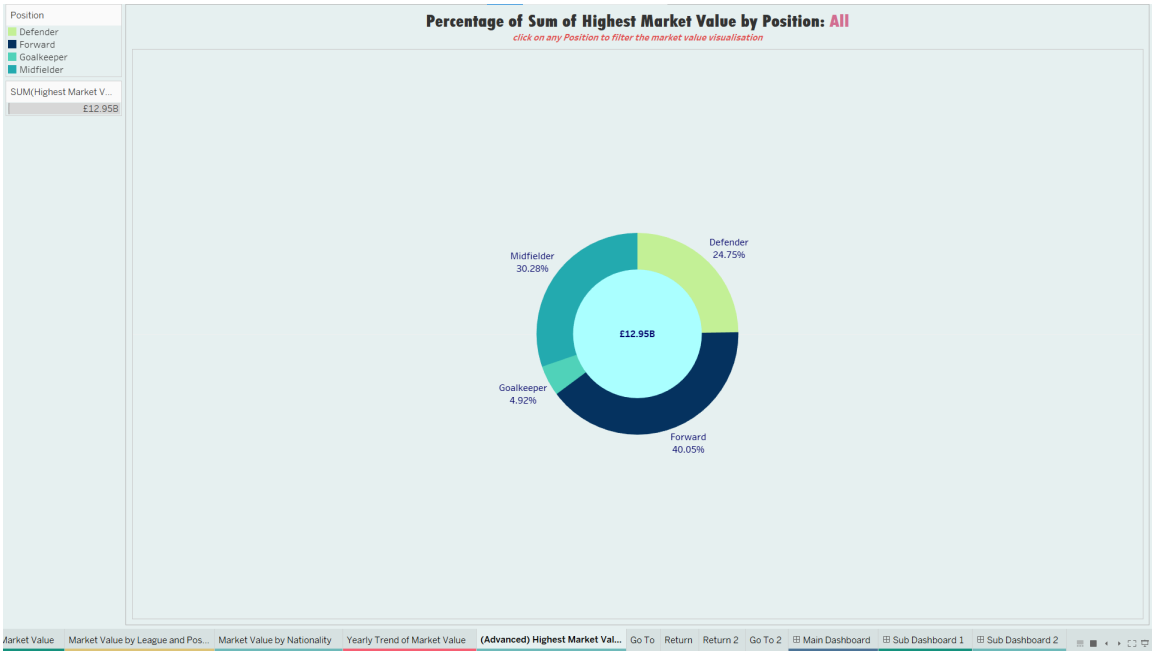
Title: Yearly Trend of Market Value

Chart Type: Heat Map

- This simple line chart shows the trend of market value over years.
- The trend follows a polynomial model with R-squared value of 0.9891.
- It is obvious that the total market values of footballers worldwide keeps increasing through the years.
- This can be explained with two main factors.
- The first factor is inflation. Worldwide economic inflation has also affected football businesses. Thus, players' prices and value keeps increasing every year, just like goods and services do.
- Second, the increase in engagement. Every year, football is watched by more and more people around the globe. To take into example, FIFA World Cup viewers have significantly increased from a total of approximately 1 Billion in 2010, to almost 5 Billion watchers in 2018. Thus, the more engagement football got through the years, the higher the overall value of footballers.
- Taking into account both factors, the trend forecast model has predicted the sum of market value to keep increasing extremely every year for the next 10 years.

- As a data analyst, this analysis needs to be updated periodically to the football management board so that they can increase their businesses to keep up with the trend.

Visualisation 7



Title: Percentage of Sum of Highest Market Value by Position: All Leagues

Chart Type: Donut Chart (Advanced Visualisation)

- The donut chart uses calculated value to show the percentage of highest market value for each position.
- The table calculation used is percent of total.
- This can be filtered by Leagues using visualisation 1.
- Based on the chart, forward takes 40.05% of the total highest market value.
- This means that in general, the forward position can reach the highest market value ever as compared to other positions.
- Just like said previously, forward is the most popular position in football.
- Players in this position are the star of their respective team, having scored the most goals and made the most assists.
- Their performance always becomes the spotlight, causing it to increase their market value.

CONCLUSION

In a nutshell, there are so many things that can be analysed and interpreted from the dataset of the transfer market value. A simple dataset can be visualised to get useful and meaningful insights to understand in deep details about a certain topic. In this context, the visualisations created are extremely useful to study further about how the transfer market behaves and what causes it to behave that way. It is also useful for football data analysts to make decisions for football clubs to buy or sell players, other than just analysing solely based on their performance on the pitch.

In general, after completing all visualisations and analysis, I am able to identify many factors that cause a player's market value to change over time other than their on field performance such as goals and assists. First, the player's league. Bigger leagues like the premier league have players with relatively high market value. Secondly, the position. We can conclude that forwards or attackers have high market values due to their role as the most important one in the team. Next, players age. Players will have higher market value during their prime years which is from 25-28 years old. After that, the player's sponsor. Players that are sponsored by bigger brands are often the best players, hence they have significantly high value compared to smaller brands. Last but not least, the nationality. We have seen that players from the United Kingdom, France, Spain and Brazil are high in value since these countries are some enormous footballing nations.

All in all, I have accomplished all the objectives of this project which is to build an advanced data visualisation dashboard in order to understand the insights in the dataset. I am also able to apply various practices in creating an interactive dashboard using certain features like actions menu, go to sheet and creating dynamic titles. To add to that, I have also completed the detailed analysis and explanation to the visualisations and dashboard created.

UNIVERSITI MALAYSIA PAHANG AL-SULTAN ABDULLAH	**SUBJECT: BSD1323 STORYTELLING AND DATA VISUALIZATION**	**MARKS**: 60(15%)
	TOPIC: CHAPTER 1 to CHAPTER 8	
	INDIVIDUAL PROJECT **DUE DATE:** 13 May 2024 - 2 June 2024	
	ID: SD23009 NAME: AHMAD AFIQ HILMY BIN HAZA HILMY SECTION: 01G	

INDIVIDUAL PROJECT: MARKING SCHEME

CLO	Description	PLO mapping	Percentage	Marks
CLO2	Demonstrate the data visualization skill using an effective storytelling.	PLO2: Cognitive Skills and Functional work skills with focus on Numeracy skills C3: Application	5%	20

CLO2 RUBRICS OF QUESTION 4

CRITERIA	LEVEL OF ACHIEVEMENT						WEIGHTAGE	SCORE
	0	1 Inadequate	2 Emerging	3 Developing	4 Good	5 Excellent		
Motivation of project topic	No motivation of the project topic provided	Very little motivation of the project topic provided	Motivation of the project topic provided but missing all major points	Motivation of the project topic provided but unclear	Clear and good motivation of the project topic provided	Very clear and excellent motivation of the project topic provided	0.5	
Details explanation of the dataset	Failed to explain the dataset	Not Efficiently, effectively, and accurately explain the dataset	Partly accurate, but not effectively explain the dataset	Effectively explain the dataset but not accurate	Accurately and effectively but not efficiently explain the dataset	Accurately effectively, and efficiently explain the dataset	0.5	
Details analysis of each dashboard	Failed to analyse the dashboards	Not Efficiently, effectively, and accurately analyse the dashboards	Partly accurate, but not effectively analyse the dashboards	Effectively analyse the dashboards	Accurately and effectively but not efficiently analyse the dashboards	Accurately effectively, and efficiently analyse each dashboard	2	
Concluding remarks	No concluding remarks provided	Very little concluding remarks provided and inaccurate	Concluding remarks provided but unclear and inaccurate	Concluding remarks provided but partly inaccurate	Clear and good concluding remarks provided	Very clear and excellent concluding remarks provided	1	
						TOTAL (20)		

CLO	Description	PLO mapping	Percentage	Marks
CLO3	Display a powerful data visualization, report, dashboard or stories in solving various applications using appropriate software.	PLO3: Functional work skills with focus on Practical, and Digital skills P4: Mechanism	10%	40

CRITERIA	LEVEL OF ACHIEVEMENT						WEIGHTAGE	SCORE
	0	1 Inadequate	2 Emerging	3 Developing	4 Good	5 Excellent		
Theory/ Knowledge on data visualization	No theoretical knowledge on data visualization observed	Very little knowledge observed on data visualization or some information is incorrect	Some knowledge or information on data visualization observed but missing all major points	Some knowledge or information on data visualization observed but still missing some major points	Good knowledge on data visualization observed, missing some minor points	Excellent knowledge on data visualization observed; provides all necessary background principles	1	
Interactive Data Visualization Techniques	Failed to demonstrate the given task	Inappropriate interactive data visualization techniques are demonstrated	Partly correct interactive data visualization techniques are demonstrated, with partly valid data	Correct interactive data visualization techniques are demonstrated, with partly valid data	Good interactive data visualization techniques are demonstrated, with valid but not completely accurate data	Competent interactive data visualization techniques are demonstrated, with valid and accurate data	1	
Theory/ Knowledge on advanced dashboard	No theoretical knowledge on advanced dashboard observed	Very little knowledge observed on advanced dashboard or some information is incorrect	Some knowledge or information on advanced dashboard observed but missing all major points	Some knowledge or information on advanced dashboard observed but still missing some major points	Good knowledge on advanced dashboard observed, missing some minor points	Excellent knowledge on advanced dashboard observed; provides all necessary background principles	1	
Advanced Dashboard Techniques & Data Validation	Failed to demonstrate the given task	Inappropriate advanced dashboard techniques are demonstrated	Partly correct advanced dashboard techniques are demonstrated, with partly valid data	Correct advanced dashboard techniques are demonstrated, with partly valid data	Good advanced dashboard techniques are demonstrated, with valid but not completely accurate data	Competent advanced dashboard techniques are demonstrated, with valid and accurate data	2	

Efficiency/ Assembly/ Tidiness	Failed to demonstrate the given task	Not efficiently, effectively and neatly demonstrated the given task	Partly efficient, but not effectively and neatly demonstrated the given task	Efficiently, but not effectively and neatly demonstrated the given task	Efficiently and effectively but not neatly demonstrated the given task	Efficiently, effectively and neatly demonstrated the given task	1	
Final Results (the advanced dashboard)	Failed to demonstrate the given task	Lack of results/ zero readability of the result. Poor originality	Partly complete result and less originality	Result presented but at low readability/ some result presented. Reader has to guess some of the missing information. Less originality	Clear, neat presentation. All required results are presented. Readability. Complete with labels, title, axes, etc.	Very Clear, neat presentation. All required results are presented. High readability. Complete with labels, title, axes, etc.	2	
						TOTAL (40)		